

Additional file 1 — Supplementary tables and figures for

*Sparse autoencoders reveal organized biological knowledge but
minimal regulatory logic in single-cell foundation models: a
comparative atlas of Geneformer and scGPT*

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This file contains supplementary tables and figures for the main manuscript.

Contents

1	Supplementary Tables	2
2	Supplementary Figures	11

1 Supplementary Tables

Table S1: **Per-ontology enrichment counts across all 18 Geneformer layers.** Each entry is the number of significant enrichments ($\text{FDR} < 0.05$) for features at that layer. GO BP = Gene Ontology Biological Process. TRRUST columns show enrichment for TF target sets (TF) and individual TF $\rightarrow$ target edges (Edges).

Layer	GO BP	KEGG	Reactome	STRING	TRRUST TF	TRRUST Edges
0	10,153	2,650	11,001	302	155	42
1	10,022	2,433	10,512	258	164	48
2	9,948	2,495	10,790	283	150	32
3	9,726	2,514	9,525	273	157	37
4	8,537	2,045	9,195	248	133	30
5	7,695	1,845	8,189	216	136	24
6	7,180	1,555	7,871	182	110	28
7	6,628	1,637	7,080	181	125	30
8	6,850	1,570	7,169	199	90	27
9	7,299	1,643	7,880	207	103	29
10	8,461	1,751	9,247	214	128	30
11	8,785	2,089	8,957	227	112	31
12	8,217	1,915	8,856	210	117	35
13	7,158	1,686	8,393	202	101	28
14	7,615	1,595	8,412	221	97	27
15	6,790	1,520	7,135	150	126	34
16	7,040	1,781	7,172	158	87	25
17	7,002	1,762	6,869	193	131	25
Total	145,106	34,486	154,253	3,924	2,222	562

Table S2: **Cross-layer feature persistence from layer 0.** Matches = features at L0 with cosine similarity > 0.7 to any feature at the target layer. The model undergoes radical representational transformation: by layer 6, essentially all features are novel with no L0 ancestry.

L0 $\rightarrow$ Target	Matches ($\cos > 0.7$)	Rate
L0 $\rightarrow$ L1	114	2.5%
L0 $\rightarrow$ L2	93	2.0%
L0 $\rightarrow$ L4	67	1.5%
L0 $\rightarrow$ L6	25	0.5%
L0 $\rightarrow$ L8	10	0.2%
L0 $\rightarrow$ L10	1	$\sim 0\%$
L0 $\rightarrow$ L12+	0	0%

Table S3: **Co-activation module statistics across all 18 layers.** PMI-based graphs with Leiden clustering (resolution = 1.0). Modules = number of distinct communities. Coverage = fraction of alive features in at least one module.

Layer	Modules	Feats in Modules	PMI Edges	Coverage
0	6	4,577	446,324	99.3%
1	8	4,562	440,681	99.0%
2	7	4,536	404,403	98.6%
3	8	4,518	393,574	98.3%
4	9	4,502	393,194	98.2%
5	12	4,472	390,845	97.7%
6	7	4,458	383,033	97.3%
7	8	4,439	371,832	96.7%
8	7	4,478	369,280	97.6%
9	7	4,535	380,304	98.7%
10	9	4,571	388,498	99.3%
11	8	4,565	388,103	99.3%
12	7	4,567	388,977	99.5%
13	7	4,561	383,779	99.5%
14	8	4,543	379,595	99.5%
15	8	4,461	340,269	98.2%
16	8	4,358	327,895	96.0%
17	7	4,474	343,059	97.7%
Total	141			

Table S4: **Top 10 causally specific SAE features at layer 11.** Δ Target and Δ Other = mean logit change at target and off-target gene positions, respectively, upon zeroing the feature. Specificity ratios were computed from unrounded values; displayed Δ values are rounded to three decimal places.

Feature	Annotation	Specificity	Δ Target	Δ Other
F2035	Cell Differentiation (neg. reg.)	$114.5\times$	-0.208	+0.002
F3692	ERAD Pathway	$108.1\times$	-0.129	-0.001
F3933	Intracellular Signaling (neg. reg.)	$55.7\times$	-0.196	-0.004
F157	Golgi Vesicle Transport	$25.4\times$	-0.056	-0.002
F3532	Protein Metabolic Process (pos. reg.)	$11.2\times$	-0.127	-0.011
F4516	Mitotic Spindle Microtubules	$10.6\times$	+0.672	+0.063
F1337	Cell Cycle Phase Transition	$9.4\times$	-0.058	-0.006
F1023	Mitotic Spindle Microtubules	$7.6\times$	-2.799	-0.367
F2936	Mitochondrion Organization	$7.1\times$	-0.366	-0.051
F3962	Endocytosis	$6.9\times$	-0.099	-0.014

Table S5: **Geneformer cross-layer information highways.** PMI between SAE feature activations at source and target layers (500K positions each). A highway = source feature with ≥ 1 target-layer feature at PMI > 3 . **Note:** the numbers here use the main-atlas 500K-position protocol; Table S18 reports the same three pairs recomputed on a 100K-position subsample with a permutation-null control. The two sets of numbers differ by less than one percentage point (e.g. L0→L5: 98.4% here vs. 98.96% in Table S18) because of the different subsample sizes, and the null correction does not change the 100K numbers.

Layer Pair	Feats w/ Deps	Mean Max PMI	Median Max PMI	Max PMI	Highways
L0 → L5	4,604	6.61	6.72	11.10	4,530 (98.4%)
L5 → L11	4,518	6.63	6.71	10.87	4,401 (97.4%)
L11 → L17	4,555	6.79	6.86	10.66	4,544 (99.8%)

Table S6: **scGPT cross-layer information highways.** Same methodology as Table S5. Note the progressive drop in downstream connectivity.

Layer Pair	PMI Edges	Upstream	Downstream	Max PMI
L0 → L4	75,305	1,935/2,027 (95.5%)	1,960/2,048 (95.7%)	9.15
L4 → L8	61,263	1,955/2,048 (95.5%)	1,723/2,048 (84.1%)	9.26
L8 → L11	45,258	1,773/2,048 (86.6%)	1,289/2,048 (62.9%)	10.78

Table S7: **Top cross-layer biological cascades.** Strongest annotated PMI connections between layer pairs. “Unlabeled” = the target feature lacks direct ontology annotation.

Pair	Source Feature	Target Feature	Biological Logic	PMI
L0→L5	Protein Processing in ER	<i>unlabeled</i>	ER stress cascade	11.10
L0→L5	mTORC1 Regulation	Autophagy	mTORC1→autophagy	9.55
L0→L5	Wnt Signaling	<i>unlabeled</i>	Wnt pathway processing	9.48
L5→L11	Protein Polyubiq. Translation	<i>unlabeled</i>	Protein quality control	10.87
L5→L11	RNA Splicing (neg. reg.)	<i>unlabeled</i>	Translational regulation	10.35
L5→L11	RNA Splicing (neg. reg.)	<i>unlabeled</i>	Post-transcriptional	10.21
L11→L17	Protein Modification	Angiogenesis (pos. reg.)	PTM→phenotype	10.62
L11→L17	COPII Vesicle Budding	Thermogenesis	Secretory→metabolic	10.29
L11→L17	Actomyosin Org.	Cell Locomotion (neg. reg.)	Structure→motility	10.14

Table S8: **Per-TF head-to-head comparison at layer 11.** Only TFs with changed specificity are shown; 40 additional TFs had no specific features in either condition.

TF	K562-SAE Specific	MT-SAE Specific	Change
ATF5	0	1	gained
BRCA1	0	1	gained
GATA1	0	1	gained
RBMX	0	1	gained
NFRKB	0	1	gained
MAX	1	0	lost
PHB2	1	0	lost
SRF	1	0	lost

Table S9: **TF feature diagnostics.** Features with known TFs in top-20 genes and TF-dominant features (≥ 3 TFs in top genes). Denominators = features with non-empty top-20 gene lists (may differ from alive counts because some features that are “dead” on the 100K held-out sample still produce gene lists from the full training data). K562-only SAE has more TF-associated features than the multi-tissue SAE.

SAE	Features with TFs in top genes	TF-dominant (≥ 3 TFs)
K562-only L11	2,967/4,598 (64.5%)	424
Multi-tissue L0	2,796/4,608 (60.7%)	452
Multi-tissue L5	2,777/4,568 (60.8%)	337
Multi-tissue L11	2,782/4,601 (60.5%)	343
Multi-tissue L17	2,680/4,603 (58.2%)	346

Table S10: **Unannotated feature analysis (original preprint).** Co-activate = unannotated feature belongs to a co-activation module containing annotated features. Isolated = no module membership. A small number of features (3 at L11, 17 at L17) belong to modules containing only unannotated features and are excluded from both columns. Clusters = standalone gene-set clusters among unannotated features. As discussed in §2.11 of the main text, the co-activate/isolated column is now regarded as weak evidence (modules cover 96–99.5% of all features so co-membership is nearly unavoidable) and is retained here only for completeness. The more informative tests appear in Table S20.

Layer	Annotated	Unannotated	Clusters	Co-activate w/ Annotated	Isolated
0	2,702	1,906	15 (48 feats)	1,876 (98.4%)	30 (1.6%)
5	2,383	2,193	19 (69 feats)	2,090 (95.3%)	103 (4.7%)
11	2,583	2,015	11 (47 feats)	1,984 (98.5%)	28 (1.4%)
17	2,154	2,426	12 (58 feats)	2,334 (96.2%)	75 (3.1%)

Table S11: **Sensitivity of the SAE-vs-SVD comparison to the cosine threshold (Geneformer, all 18 layers).** For each layer and each cosine threshold $\tau \in \{0.3, 0.5, 0.7, 0.9\}$: percentage of alive SAE features whose decoder column has maximum absolute cosine above τ with any of the top-50 SVD axes (columns “aligned%”), and percentage of ontology enrichments found exclusively in the non-aligned (“novel”) set (columns “%enrich novel”). The value $\tau=0.5$ reproduces the per-layer SVD-aligned counts in Table 1 (the original preprint quoted $\tau=0.7$ in the text but the pipeline actually used $\tau=0.5$; we have corrected the main text). Across the full range, $> 98\%$ of ontology enrichments live exclusively in the novel set, so the headline conclusion is robust.

Layer	Aligned %				% enrich in novel			
	$\tau=0.3$	$\tau=0.5$	$\tau=0.7$	$\tau=0.9$	$\tau=0.3$	$\tau=0.5$	$\tau=0.7$	$\tau=0.9$
0	4.19	0.89	0.02	0.00	98.88	99.98	100.00	100.00
1	3.45	0.59	0.07	0.00	99.54	99.98	100.00	100.00
2	3.73	0.63	0.04	0.00	99.49	99.93	100.00	100.00
3	3.26	0.26	0.00	0.00	99.27	100.00	100.00	100.00
4	2.69	0.28	0.00	0.00	99.48	99.97	100.00	100.00
5	2.63	0.17	0.00	0.00	98.92	100.00	100.00	100.00
6	2.39	0.11	0.00	0.00	98.62	99.99	100.00	100.00
7	2.08	0.13	0.00	0.00	98.30	99.99	100.00	100.00
8	1.48	0.07	0.00	0.00	99.40	99.98	100.00	100.00
9	1.09	0.09	0.00	0.00	99.36	99.99	100.00	100.00
10	1.09	0.15	0.00	0.00	99.63	100.00	100.00	100.00
11	0.74	0.09	0.00	0.00	99.28	100.00	100.00	100.00
12	0.65	0.07	0.00	0.00	99.26	100.00	100.00	100.00
13	0.95	0.07	0.00	0.00	98.75	100.00	100.00	100.00
14	0.93	0.04	0.00	0.00	97.90	99.55	100.00	100.00
15	1.09	0.11	0.00	0.00	98.28	99.03	100.00	100.00
16	1.00	0.11	0.02	0.00	98.73	99.02	99.47	100.00
17	1.63	0.26	0.02	0.00	98.68	99.99	100.00	100.00

Table S12: **PMI co-activation graph dimensions and peak memory (Geneformer layers 0 and 11).** Recomputed on a deterministic 500,000-position subsample (seed 42) of the 4,056,351 K562 token positions at each layer, with MIN_COACTIVATION=20 and PMI_THRESHOLD=2.0 to match the main-text co-activation pipeline (main-text numbers use the full 4M positions with MIN_COACTIVATION=50). N_{feat} = alive features at the layer. Peak RSS measured with `psutil.Process().memory_info().rss`. The dense ($N \times N$) coactivation matrix accounts for ≈ 85 MB of the budget; the remaining ~ 700 MB is the subsampled TopK index buffer and PyTorch/Python overhead.

Layer	N_{feat}	Subsampled positions	$ E $ (after thresh.)	Coact matrix (MB)	Peak RSS (MB)	Encode (s)	PMI (s)
0	4,608	500,000	435,252	85	811	100.1	11.0
11	4,608	500,000	356,894	85	864	93.6	10.3

Table S13: **Leiden resolution sweep at Geneformer layers 0 and 11.** Module count (≥ 3 features), mean module size, coverage (fraction of graph nodes in ≥ 3 -feature modules), and Adjusted Rand Index vs. the baseline $\gamma=1.0$ partition for $\gamma \in \{0.5, 0.75, 1.0, 1.5, 2.0\}$. **Note on comparison with Table S3:** this sweep is run on a deterministic 500K-position subsample with MIN_COACTIVATION=20 (see caption of Table S12), whereas the main-text module counts in Table S3 use the full 4,056,351 positions with MIN_COACTIVATION=50. The different PMI significance thresholds produce slightly different partition sizes; at $\gamma=1.0$ the sweep gives 8 modules at L0 and 9 at L11, versus 6 and 8 in Table S3. This disagreement is expected given the parameter change and should not be read as instability. The partition is *not* stable in a strong ARI sense (ARI between $\gamma=1.0$ and its immediate neighbors is in the 0.17–0.26 range); instead, the co-activation graph exhibits a nested hierarchy in which low γ collapses the graph into ~ 2 very large communities and high γ fragments each main community into finer subcommunities. The qualitative biological identities of modules (cell cycle, translation, metabolism, etc.) are preserved across the range; our main-text module counts correspond to the resolution-1.0 cut of this hierarchy.

Layer	γ	Modules (≥ 3)	Mean size	Coverage	ARI vs $\gamma=1.0$
0	0.50	2	2,287.5	100.0%	+0.027
0	0.75	3	1,525.0	100.0%	+0.259
0	1.00	8	571.9	100.0%	+1.000
0	1.50	25	183.0	100.0%	+0.252
0	2.00	48	95.3	100.0%	+0.117
11	0.50	2	2,282.0	100.0%	+0.013
11	0.75	3	1,521.3	100.0%	+0.180
11	1.00	9	507.1	100.0%	+1.000
11	1.50	31	147.2	100.0%	+0.171
11	2.00	56	81.5	100.0%	+0.085

Table S14: **SAE hyperparameter ablation at Geneformer layer 11.** Grid over expansion ratio $\in \{2\times, 4\times, 8\times\}$ and TopK sparsity $k \in \{16, 32, 64\}$, trained on the same 1M-position K562 subsample used in the main atlas (Methods), evaluated on 100K held-out positions. Columns: expansion, k , total features, variance explained on held-out positions, number of dead features (no activation on the 100K hold-out), mean absolute decoder–decoder cosine similarity on a 5,000-pair sample (a proxy for quasi-orthogonality), and Leiden module count (≥ 3 features, $\gamma=1.0$) on a 200K-position co-activation graph. The baseline configuration ($4\times, k=32$) sits at the center of the grid and does not produce an unusual point: variance explained increases monotonically with both expansion and k , module count is within 7–12 across the entire grid, and quasi-orthogonality is preserved everywhere ($\langle |\cos| \rangle \lesssim 0.05$). The main-text qualitative conclusions—substantial superposition, organized biological structure, modular co-activation—are preserved across the grid. We report the structural metrics here; full downstream annotation rate and TF specificity per grid cell are beyond this revision’s compute budget (see §2.12).

Expansion	k	N_{feat}	VarExpl	Dead	$\langle \cos \rangle$	Modules (≥ 3)
$2\times$	16	2,304	0.706	95	0.0421	9
$2\times$	32	2,304	0.752	11	0.0383	10
$2\times$	64	2,304	0.795	1	0.0338	8
$4\times$	16	4,608	0.790	228	0.0394	12
$4\times$	32	4,608	0.819	45	0.0365	7
$4\times$	64	4,608	0.846	1	0.0327	7
$8\times$	16	9,216	0.846	559	0.0360	12
$8\times$	32	9,216	0.866	112	0.0345	8
$8\times$	64	9,216	0.884	8	0.0329	6

Bold row: main-atlas baseline used in Table 1 and throughout the paper.

Table S15: **scGPT perturbation response mapping on Replogle K562 CRISPRi, layer 7.** 50 perturbation targets (top of the same target list used in the Geneformer Table 6 test), 10 cells per target, 100 non-targeting K562 control cells, per-cell binned expression values re-tokenized from the Replogle source h5ad. Responding features = mean activation effect size vs. control $|z| > 0.5$. “Specific” definitions match Table S16. Both TRRUST and DoRothEA A+B+C yield single-digit loose overlap rates, and zero FDR-corrected specific TFs—in the same qualitative regime as the Geneformer L11 numbers in Table S16.

Database	DB TFs	In panel	Loose n	Loose rate	FDR<0.05 n	FDR rate
TRRUST	795	28	2	7.14%	0	0.00%
DoRothEA (A+B+C)	429	12	2	16.67%	0	0.00%

Table S16: **Perturbation specificity across regulatory databases.** Same experimental protocol as Table 6 of the main text (Geneformer L11 SAE, Replogle CRISPRi, 100 perturbation targets) but tested against three regulatory priors: TRRUST (Han et al., 2018; literature-mined), DoRothEA A+B+C (Garcia-Alonso et al., 2019; high-confidence subset), and DoRothEA all (all confidence levels). Columns: database, total TFs in the database, TFs present in the Replogle panel with ≥ 5 known targets, number of TFs with at least one responding feature whose top-20 gene list overlaps a known target (“loose”), loose rate, number with a significant hypergeometric enrichment of their known targets in the union top-20 gene list across responding features (BH-corrected FDR<0.05 across the TF panel, universe = 20,000 human protein-coding genes), and FDR rate. The “loose” column climbs with database size because larger target sets make coincidental overlaps more likely; the FDR-corrected column, which controls for target-set size via the hypergeometric test, stays in the single-digit range across all three databases. Note: the loose TRRUST rate of 3.57% here (1/28) applies the same ≥ 5 -known-targets cutoff used for DoRothEA; the 6.2% headline rate in Table 6 of the main text uses the main-text pipeline’s feature-level definition of “specific response” and a 48-TF panel with no ≥ 5 cutoff.

Database	DB TFs	In panel	Loose n	Loose rate	FDR<0.05 n	FDR rate
TRRUST	795	28	1	3.57%	0	0.00%
DoRothEA (A+B+C)	429	12	4	33.33%	1	8.33%
DoRothEA (all)	643	18	7	38.89%	1	5.56%

Table S17: **scGPT causal patching at layer 7: proxy (uniform 1.0) vs. true binned expression values.** Lean head-to-head on a 20-cell $\times$ 10-feature subset drawn from the Tabula Sapiens extraction panel, with features re-tokenized from the source h5ad files (`tabula_sapiens_{immune,kidney,lung}.h5ad`) to recover the per-cell binned values. Specificity is measured as a simplified layer-7 SAE reconstruction delta after single-feature ablation (not final-layer logit change), so the absolute numbers are not directly comparable to the $0.98\times$ figure in §2.7 of the main text; the comparison between the two modes is, however, controlled. Results are essentially identical between the two input regimes, indicating that the preprint’s low scGPT causal specificity is *not* an artifact of the uniform-value proxy.

Mode	Median specificity	Mean specificity	Max specificity	# features $> 2\times$
Proxy (uniform 1.0)	0.000	1.657	7.062	3
True (per-cell bins)	0.000	1.631	6.730	3

Table S18: **Null-corrected cross-layer highway fractions for the three long-range Geneformer pairs.** 100K gene positions subsampled from the K562 activation cache, encoded through the per-layer SAEs, cross-layer PMI computed as in Methods. Columns: observed 99th percentile of cross-layer PMI, permutation-null 99th percentile (τ_{null}^{p99} , max across 20 column-shuffle rounds), effective threshold $\tau = \max(3.0, \tau_{\text{null}}^{p99})$, and highway fraction. The permutation null’s 99th-percentile PMI is *below* 3.0 in every pair tested, so the fixed threshold $\tau=3.0$ is already conservative and the null correction leaves the highway fraction unchanged. This argues that the $\sim 97\text{--}99\%$ highway rate is not an artifact of the $k=32$ TopK sparsity.

Src	Tgt	Obs $p99$ PMI	Null $p99$ PMI	τ_{eff}	Fixed- τ highway%	Null-corr highway%
L0	L5	4.76	2.46	3.00	98.96%	98.96%
L5	L11	4.93	2.76	3.00	96.66%	96.66%
L11	L17	4.77	2.85	3.00	99.02%	99.02%

Table S19: **Consecutive-layer highway fractions $L_i \rightarrow L_{i+1}$ (Geneformer, fixed- τ).** Same protocol as Table S18 (100K gene positions per layer, fixed $\tau=3.0$). Because the permutation null’s 99th percentile PMI is below 3.0 for every long-range pair tested in Table S18, we do not re-run the permutation null on every consecutive pair. The curve is near-flat in the 94.7–99.1% range with no sharp drop at any single transition, supporting continuous cross-layer information flow rather than concentrated bottlenecks. The slight dip at the final layer pair (L16→L17) is consistent with the terminal-layer specialization reported in §2.4.

$L_i \rightarrow L_{i+1}$	Highway%
L0 → L1	98.68%
L1 → L2	98.74%
L2 → L3	98.48%
L3 → L4	98.35%
L4 → L5	97.70%
L5 → L6	97.09%
L6 → L7	97.55%
L7 → L8	96.96%
L8 → L9	97.37%
L9 → L10	98.24%
L10 → L11	99.13%
L11 → L12	99.05%
L12 → L13	99.05%
L13 → L14	98.87%
L14 → L15	98.48%
L15 → L16	96.72%
L16 → L17	94.70%

Table S20: **Cell-type and perturbation-response tests for unannotated features (Geneformer layers 0, 5, 11, 17).** Columns: #unann = number of features with zero ontology enrichments; CT% unann = fraction of those that are enriched for ≥ 1 of the 56 Tabula Sapiens cell types (Fisher’s exact, BH FDR<0.05); CT% ann = same for the annotated population; CT% null = expected fraction for a matched random subset of size #unann drawn from all alive features; Pert% unann = fraction of unannotated features that appear in the top-20 responding list for ≥ 1 of the 100 Replogle CRISPRi targets with $|\text{effect}| > 0.5$; Pert% ann and Pert% null defined analogously. Cell-type rates are uniformly high ($\approx 90\text{--}95\%$) for both populations, reflecting that nearly every SAE feature at these layers is cell-type-selective in Tabula Sapiens; this test does not separate the two populations cleanly. The perturbation test does separate them: at layer 11 unannotated features are $\sim 15\times$ more likely than annotated features to appear in a top-20 responding list, providing non-tautological evidence that a subset of unannotated features carry real perturbation-response signal.

Layer	#unann	CT% unann	CT% ann	CT% null	Pert% unann	Pert% ann	Pert% null
0	1,906	94.96%	94.52%	94.70%	0.42%	0.22%	0.30%
5	2,225	91.73%	95.80%	93.85%	2.20%	1.72%	1.93%
11	2,025	92.94%	95.74%	94.53%	2.32%	0.15%	1.11%
17	2,454	89.41%	94.89%	91.96%	3.63%	3.81%	3.70%

Table S21: **Matched-input Geneformer variance explained at Tabula Sapiens layers 0, 5, 11.** K562-trained Geneformer SAEs applied to the same 3,000 Tabula Sapiens cells used for the scGPT atlas (L0, L5, L11 activations already extracted in the phase-3 pipeline). 200K position subsample per layer, variance explained computed on held-out batches without retraining. When Geneformer SAEs see out-of-distribution data (Tabula Sapiens, not K562), variance explained drops by 18–23 percentage points, implying that the 9-percentage-point Geneformer-vs-scGPT variance-explained gap in Table 3 of the main text is a lower bound on the architectural gap in Tabula-Sapiens-like distributions: under matched input, the gap widens to ~ 27 pp (e.g. 0.636 Geneformer vs. scGPT’s 0.886 mean VarExpl at comparable depth). Alive feature counts are essentially unchanged between the two input distributions, meaning this is not a case of SAEs collapsing to a subset of dead units.

Layer	K562 VarExpl	K562 alive	TS-matched VarExpl	TS-matched alive	$\Delta\text{VarExpl}$
0	0.846	4,606	0.618	4,608	−0.228
5	0.854	4,537	0.638	4,539	−0.216
11	0.820	4,577	0.636	4,575	−0.184

Table S22: **scGPT feature mutual-information with tissue vs. cell type.** 150K-position subsample at each of scGPT layers {0, 4, 7, 11}. “Tissue-dominant” = feature whose per-feature MI with tissue (3-way: immune/kidney/lung) exceeds its MI with cell type (56-way). Tissue here is a proxy for sample-level batch because the cached scGPT extraction metadata records only {tissue, cell_type}, not donor ID or sequencing assay. Across all four layers, mean MI with cell type is $\sim 2.5\times$ larger than mean MI with tissue, and essentially zero features are tissue-dominant. A full donor-level analysis would require joining the Tabula Sapiens source h5ad on cell_idx; we flag this as a follow-up.

Layer	Alive feats	Mean MI(tissue)	Mean MI(cell_type)	Tissue-dominant	Frac tissue-dominant
0	1,990	0.0003	0.0008	0	0.00%
4	2,038	0.0005	0.0015	0	0.00%
7	1,999	0.0009	0.0023	0	0.00%
11	1,766	0.0016	0.0040	1	0.06%

2 Supplementary Figures

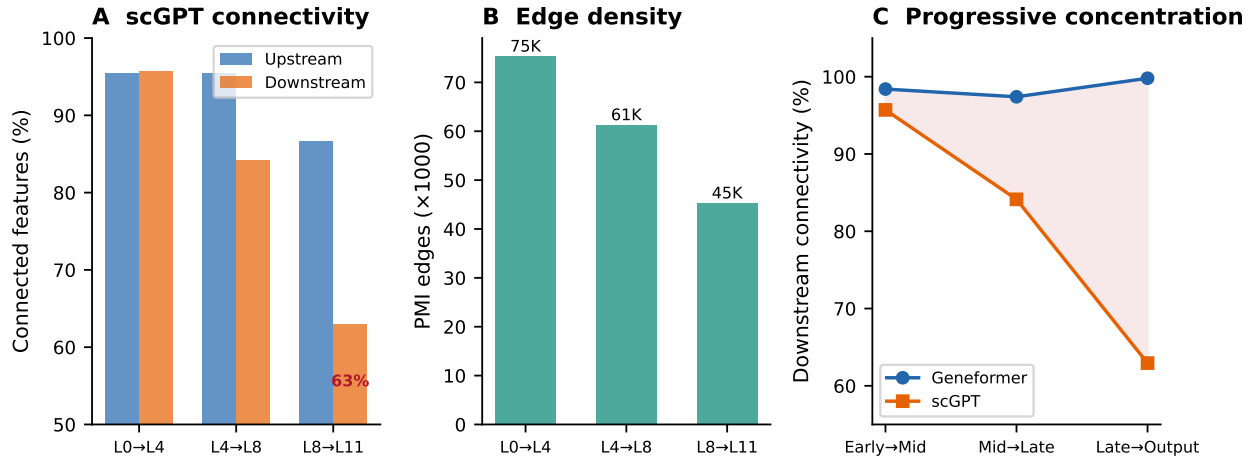


Figure S1: **scGPT cross-layer connectivity reveals progressive information concentration.** (A) Upstream connectivity remains high (86–96%) but downstream connectivity drops sharply from 96% to 63% across layer pairs, indicating progressive bottlenecks. (B) PMI edge density decreases from 75K to 45K edges across the three layer pairs. (C) Comparison with Geneformer: Geneformer maintains near-complete downstream connectivity (97–100%) while scGPT drops to 63%, suggesting fundamentally different information flow architectures.

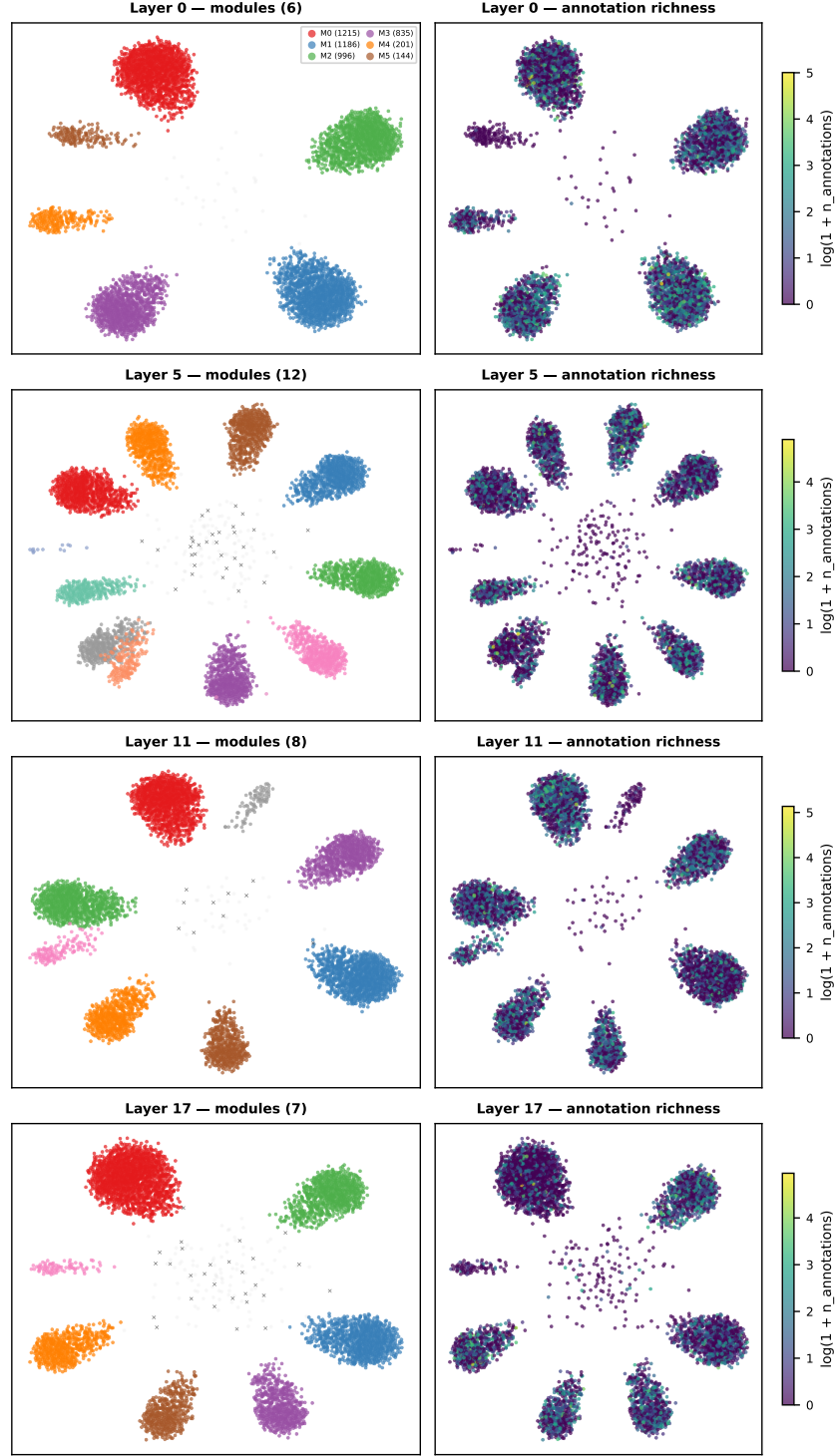


Figure S2: **Co-activation network layout of SAE features across layers.** All panels are **force-directed (Fruchterman–Reingold)** layouts of the **intra-module co-activation graph** produced with NetworkX; they are *not* UMAP or t-SNE projections of the feature decoder vectors (which are quasi-orthogonal by construction and therefore produce structureless UMAP point clouds). Left column: nodes colored by Leiden module membership. Each module forms a spatially distinct community. Right column: same layouts colored by annotation richness (log-transformed number of significant enrichment terms). Unassigned features (gray) cluster centrally. Module count varies from 6 (L0) to 12 (L5), reflecting the different co-activation patterns at each layer.

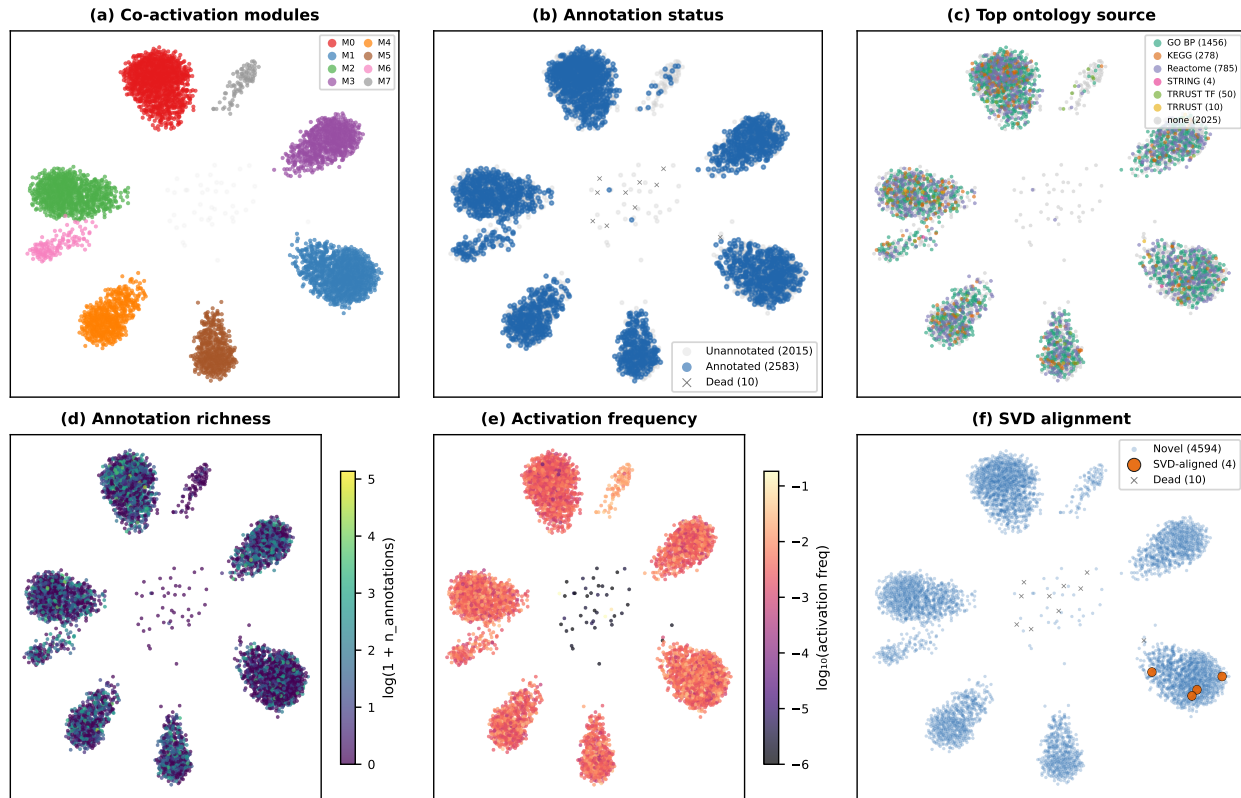


Figure S3: **Six-panel force-directed co-activation layout of layer 11 SAE features.** All six panels share the same **Fruchterman–Reingold (NetworkX) layout of the intra-module co-activation graph**; the panels differ only in the node-coloring overlay. This is *not* a UMAP, t-SNE, or PCA projection of decoder weight vectors. **(a)** Eight Leiden modules form spatially distinct communities. **(b)** Annotated features (blue) distribute across all module clusters; unannotated features (gray) concentrate centrally. **(c)** Top ontology source reveals module-specific enrichment patterns: certain modules are dominated by GO BP (green), others by STRING interactions (pink) or Reactome pathways (purple). **(d)** Annotation richness gradient across modules. **(e)** Activation frequency varies across modules. **(f)** SVD-aligned features (orange, $n = 4$) are scattered across different modules, while 4,594 novel features (blue) fill the landscape.

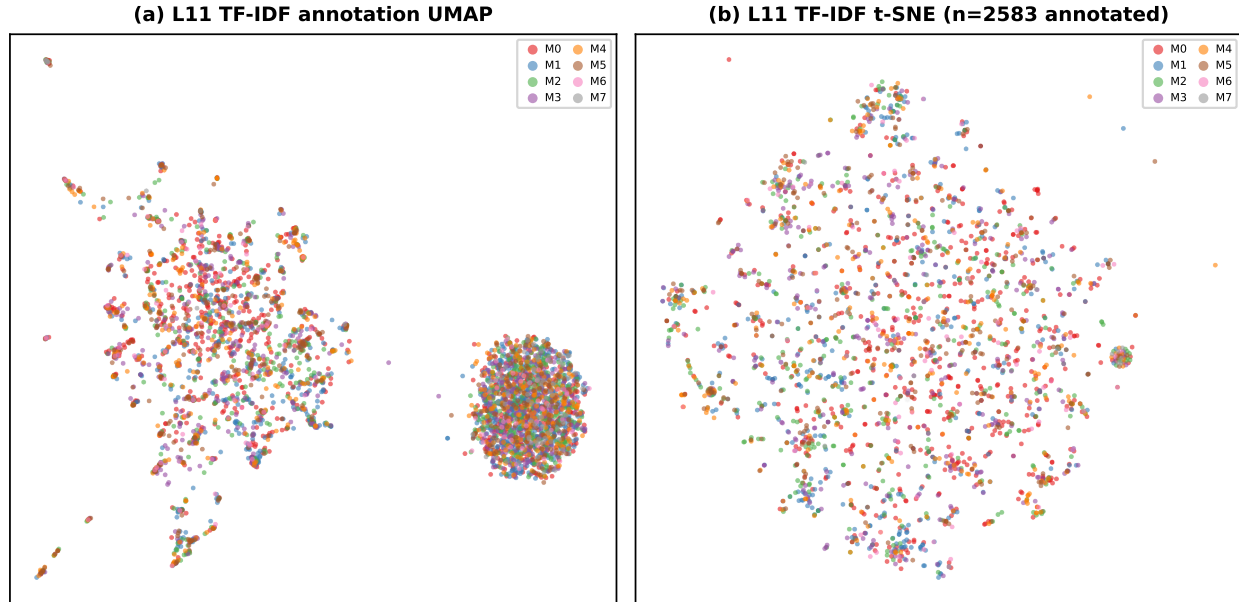


Figure S4: **Annotation-based projections provide independent validation.** **(a)** UMAP of TF-IDF weighted ontology term vectors for layer 11 (4,608 features). Annotated features (left cluster) separate from unannotated features (right blob), with internal structure reflecting shared biological annotations. **(b)** t-SNE of TF-IDF annotation vectors for annotated features only ($n = 2,583$). Fine-grained subclusters partially correspond to co-activation modules, confirming that module structure reflects genuine biological similarity.